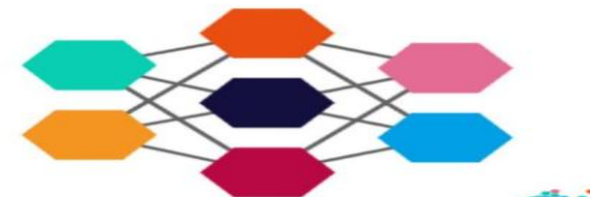




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Topology Meets the Brain

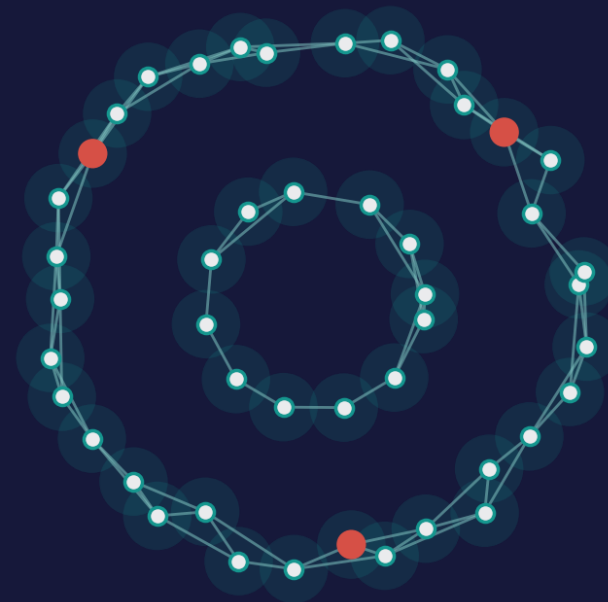
Topological Machine Learning for Affective and Cognitive State Decoding in Brain–Computer Interfaces

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Why decoding affect and cognition is hard

1

Non-stationarity

Neural dynamics drift within a session; a model fitted at minute 2 degrades by minute 20.

2

Low SNR, high artifact

Scalp EEG is μ V-scale and shares the recording with ocular, muscular and line noise.

3

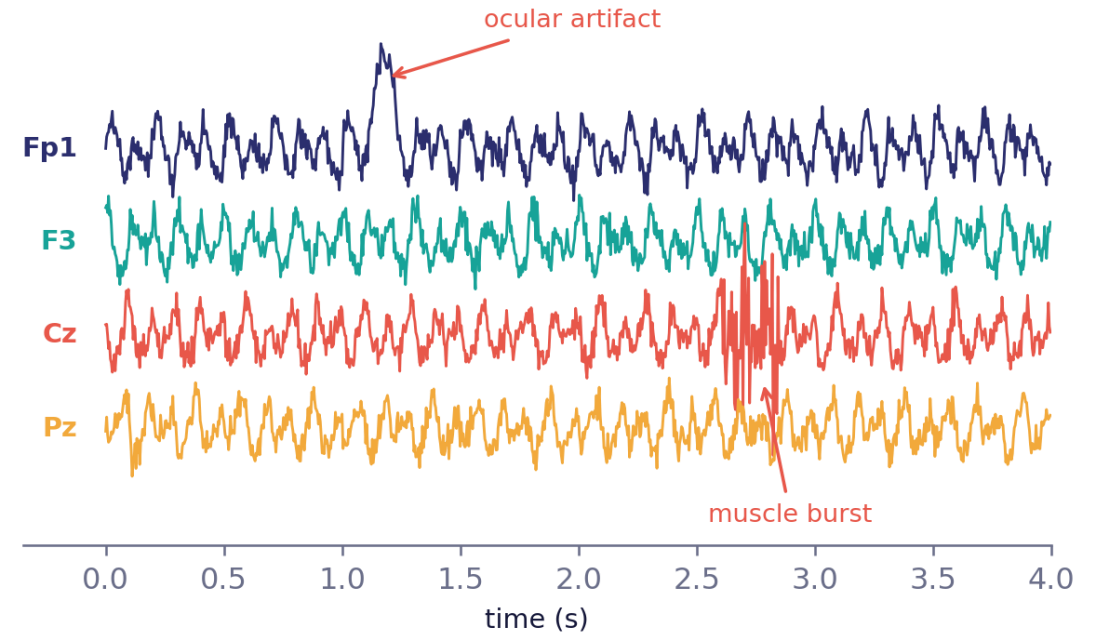
Inter-subject variability

Anatomy and electrode placement shift spectral features; cross-subject transfer collapses.

4

Entangled states

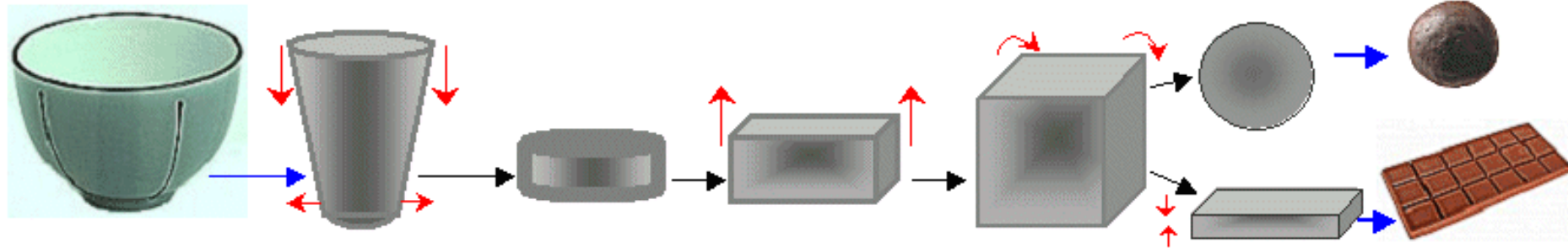
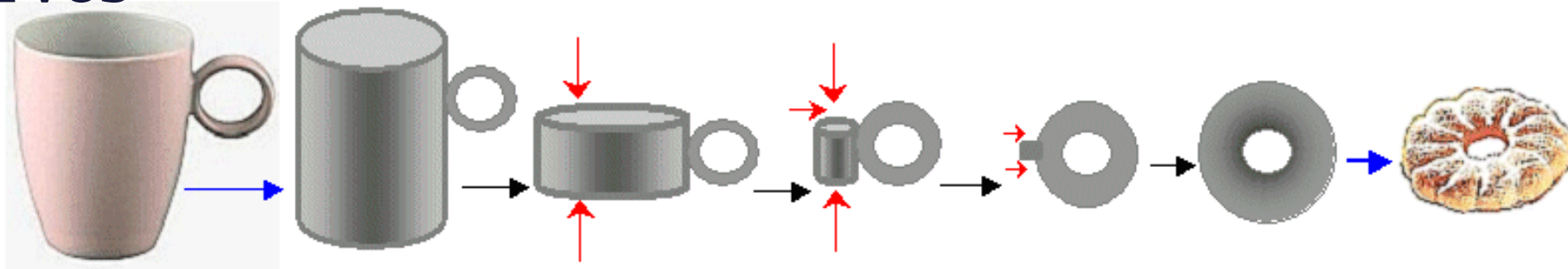
Affect and cognitive load co-vary — arousal raises theta, load blunts affective response.



Spectral features encode how much energy sits in a band — not how the state is organised in time.



Shape without measurement: what topology actually preserves



Deform, don't cut

Stretching, squashing and bending are allowed. Cutting and gluing are not — so a mug becomes a doughnut.

Holes are the invariant

Both rows preserve one property throughout: the mug and doughnut keep one hole; the bowl and sphere keep none.

Why the brain cares

Neural trajectories can be reshaped by amplitude or noise yet keep the same topology — the signature we decode.

Data has shape — and shape is measurable

Topology studies the properties of a space that survive continuous deformation: stretching, bending and rescaling. Two brain states may have the same mean and variance and still be organised differently in state space.

Topological data analysis (TDA) turns that organisation into numbers a classifier can use.

 β_0

Connected components

How many separate clusters the dynamics visit

 β_1

Loops / cycles

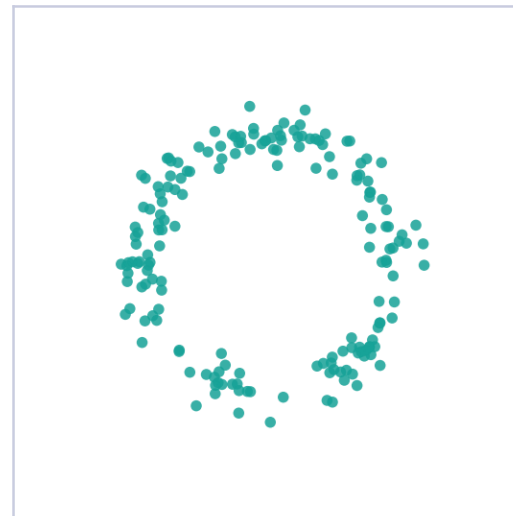
Recurrent, quasi-periodic orbits — rhythmic activity

 β_2

Voids / cavities

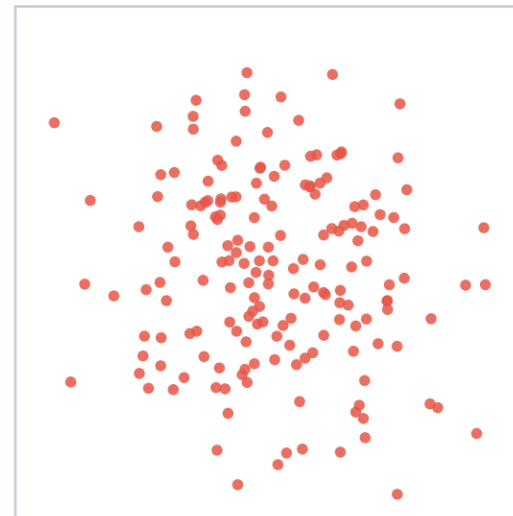
Higher-order coupling across channels

Structured state



$$\beta_0 = 1 \quad \beta_1 = 1$$

Unstructured state



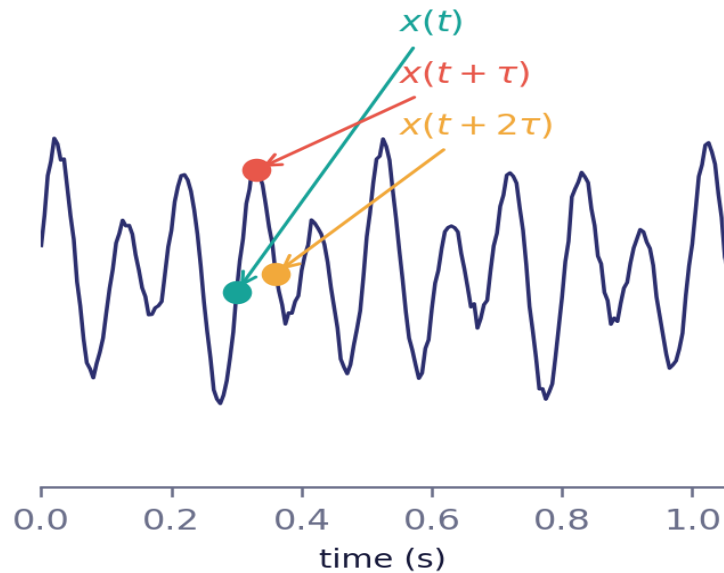
$$\beta_0 = 1 \quad \beta_1 = 0$$

Same mean, same variance — different shape

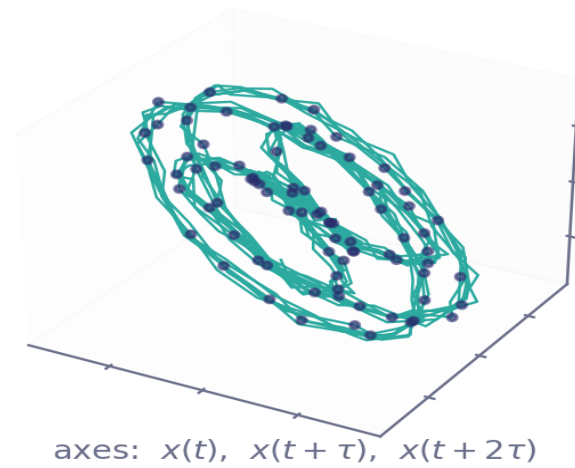
Same first- and second-order statistics. Different Betti numbers. That difference is the feature.

Takens' delay embedding

Single EEG channel



Delay-embedded attractor



Delay-coordinate map

$$v(t) = [x(t), x(t+\tau), \dots, x(t+(m-1)\tau)]$$

Takens' theorem: for a generic observable, the delay embedding is diffeomorphic to the true attractor — its topology is preserved.

One channel, one epoch → a cloud of points.

● τ · time delay

First minimum of the mutual information curve (or first zero of autocorrelation)

● m · embedding dimension

Smallest m at which false nearest neighbours vanish; $m = 3-10$ for EEG

● Window length

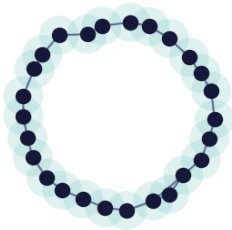
2–4 s epochs: long enough for the attractor to be sampled, short enough to stay stationary

Persistent homology: grow, watch, record

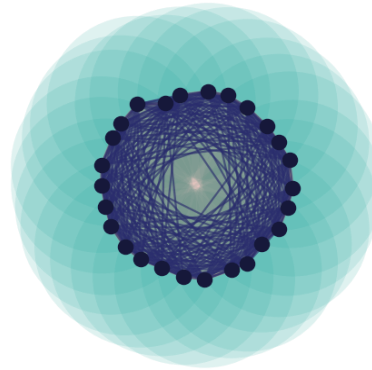
ϵ small
26 components, no loop



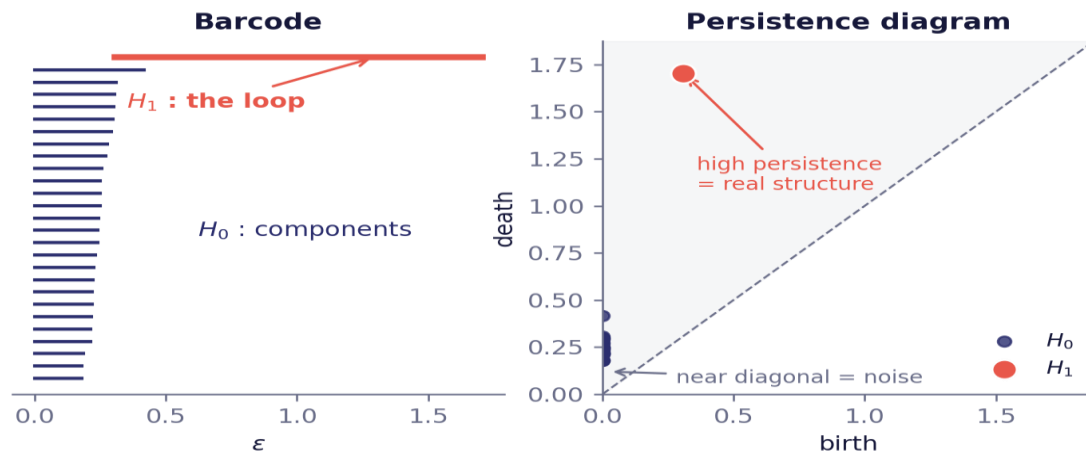
ϵ medium
1 component, loop born



ϵ large
loop dies, disk fills



Vietoris–Rips filtration: place a ball of radius ϵ on every point and grow ϵ from 0 to ∞



Every feature gets a birth and a death.

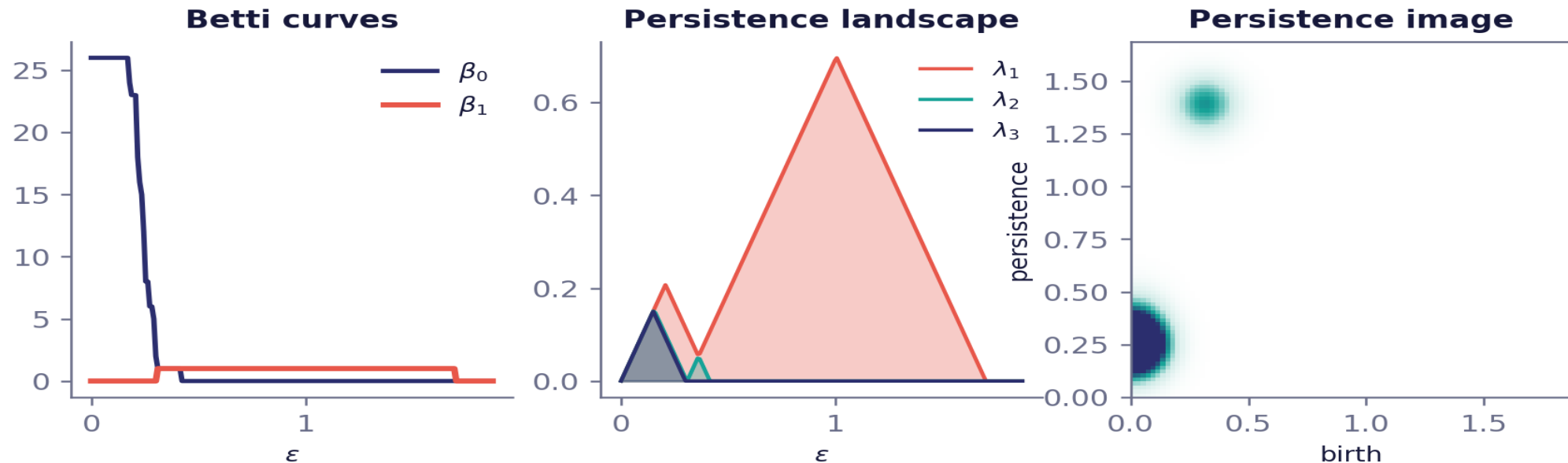
Persistence = death – birth. Long bars are real structure; short bars hugging the diagonal are noise. The diagram is a denoised, multi-scale summary of the whole epoch.

Why this matters for EEG

- No threshold, no band choice — every scale is examined at once
- H_0 tracks how fragmented the dynamics are
- H_1 tracks recurrent, quasi-periodic orbits

Persistence diagrams are not feature vectors — yet

Diagrams live in a metric space with no inner product and no fixed length. Vectorisation maps them into $\mathbb{R}^n$ so that any standard classifier can consume them.



● Betti curves

β_k as a function of ϵ — cheap, interpretable, fixed length

● Landscapes

Stable, invertible, live in a Hilbert space — kernel-friendly

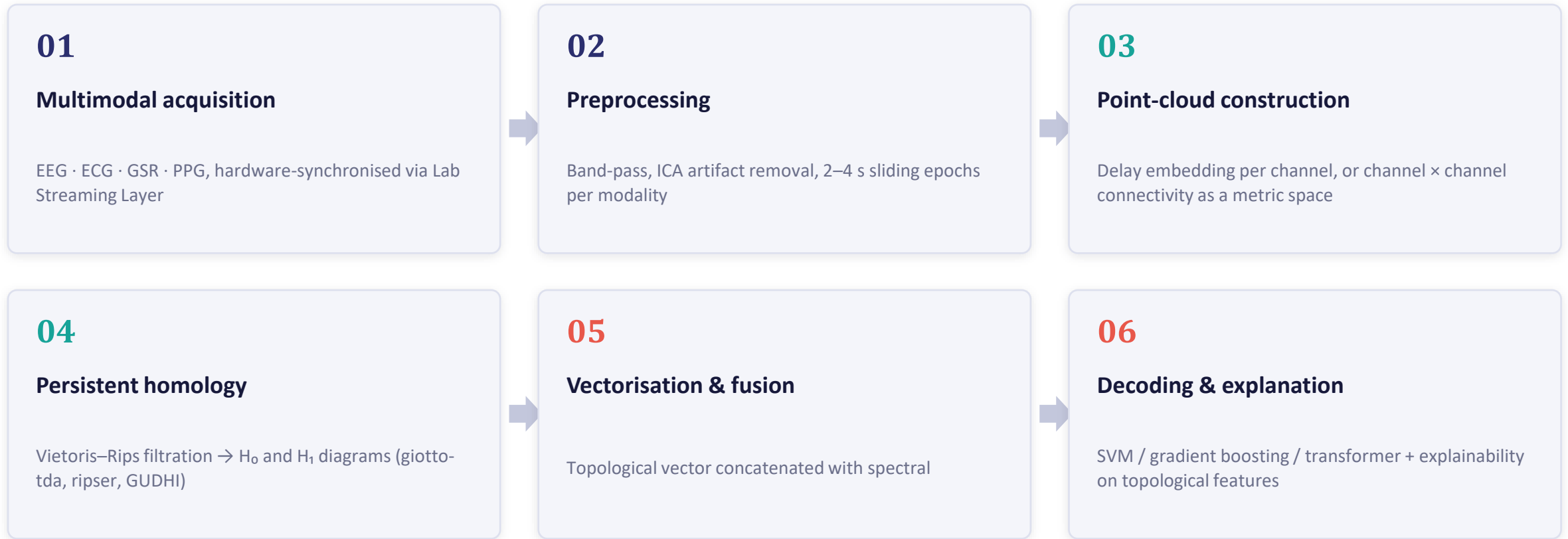
● Persistence images

Fixed-size raster $\rightarrow$ direct input to a CNN

● Scalar summaries

Persistent entropy, total persistence, max lifetime

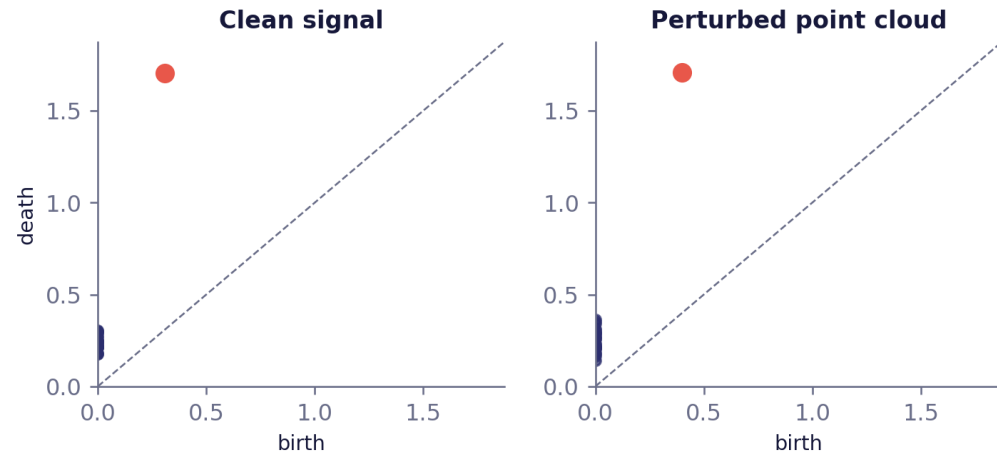
An end-to-end topological BCI pipeline



Output: joint estimate of affective state (valence × arousal) × cognitive workload — continuous, per-epoch, and attributable to specific topological features



What topology buys — and what it costs



The stability theorem

A bounded perturbation of the data produces a bounded change in the persistence diagram: the bottleneck distance between diagrams is at most the perturbation of the underlying function. Noise cannot manufacture a long bar.

This is the formal reason topological features are robust to EEG artifacts.

Strengths

- **Invariance** — robust to amplitude scaling, rotation and monotone reparametrisation
- **Provable stability** — small data perturbation $\Rightarrow$ small diagram change
- **Interpretability** — a long H_1 bar is a recurrent orbit, not an opaque activation
- **Small-sample friendly** — few parameters; works where deep nets overfit
- **Modality-agnostic** — the same pipeline runs on EEG, ECG, GSR and PPG

Open challenges

- **Cost** — Vietoris–Rips scales steeply with point count — subsampling or landmarks needed
- **Parameter sensitivity** — results depend on τ , m and window length
- **Phase blindness** — topology discards information a spectral method retains
- **Benchmarking gap** — no shared protocol for comparing TDA-BCI results



WHAT COMES NEXT

01

Multimodal topological fusion

Joint point clouds across EEG, ECG, GSR and PPG rather than four independent pipelines

02

Joint affect × workload decoding

An emotional n-back paradigm: orthogonal manipulation of load (0–3 back) and IAPS-normed affective content

03

Closing the dataset gap

No public corpus yet pairs EEG+ECG+GSR+PPG with continuous joint affect and workload labels

04

Real-time and explainable

Landmark-based persistence for online decoding, with SHAP attribution over topological features

If the brain's states differ in shape, we should be measuring shape.

Thank you — questions welcome

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